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| **PI Name:** | Fornace, Albert M.D. | | |
| **Project Title:** | NSCOR: Space Radiation and Intestinal Tumorigenesis: Risk Assessment and Counter Measure Development | | |
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| **Key Personnel Changes/Previous PI:** | | | |
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**Performance Goal Text:**

**Task Description:**

The overall goal of this project is to improve NASA risk estimates for space radiation-induced intestinal tumors. We are employing APC mouse models that have been successfully used to demonstrate radiation-induced intestinal cancer. The APC models have inactivation of one APC (adenomatous polyposis coli) gene and the mice develop intestinal tumors, adenomas. Radiation can inactivate the remaining APC gene and thus initiate increased intestinal tumor development. For our NSCOR study we are employing APC1638N model, which was generated by a targeted modification of the Apc gene at a position corresponding to aa 1638. These mice develop very few adenomas in small as well as in large intestine but post-radiation tumor numbers shows marked increases. The CDX2P Apc flox/+ the other APC mouse model we will be employing has one APC allele inactivated almost exclusively in the colon leading to adenomas and subsequent adenocarcinomas in the colon only. Data from these two mouse models will be combined with our previous studies on ApcMin/+ mice, a widely studied model for colorectal cancer to calculate risk estimates for NASA.

The strategy is to quantitatively analyze intestinal tumor incidence and grade following exposure to HZE ions, and/or high-energy protons of various energies, LET, and doses, and compare to gamma rays in the APC1638N and CDX2P Apc flox/+ models. Based on results in the Dr. Shay's laboratory, efforts will also include protracted irradiation either with small graded doses given daily over 1 to 2 weeks or low dose rate, when available. We have initiated experiments to assess potential gender and age variation in tumor incidence with space radiation and also to characterize the effects of space radiation in triggering persistent stress responses and their potential effects on tumorigenesis. The plan is also to assess other mouse models proposed in the project to determine the relative effects of space radiation at other points in the carcinogenic process. In the future efforts will be put forward to delineate cancer progression-specific biological changes and genomic, proteomic and metabolomic signatures associated with radiation-induced tumorigenesis and perturbations in tumor suppressor and oncogene-related pathways in radiation-induced tumors.

Since low-LET risk estimates are available for human populations, we will then be able to model the relative risk for space radiation. In particular, this ratio can then be extrapolated to human exposure to low-LET radiation to then make an estimate of risk for colorectal cancer in humans exposed to space radiation.

In order to answer the critical question of risk for colonic tumorigenesis after acute and chronic space radiation exposures, adequate in vitro human cell culture and animal models for CRC initiation and progression are needed. We propose to better understand the effects of high-LET particle and high-energy protons in colonic epithelial cells with respect to the nature of damage and consequent tumor induction using biological cancer endpoints and systems biology (genomics, proteomics and metabolomics) approaches. In addition, we propose to compare these results to the biological consequences of low dose rates and mixed fields of radiation which astronauts are likely to be continuously exposed. The study of low- and high-LET radiation on colonocytes gains more relevance with increasing age when premalignant adenomas and other colonic lesions such as flat lesions and microscopic aberrant crypt foci become more prevalent. Pre-existing undetectable genetic or chromosomal aberrations in astronauts in addition to de novo mutations resulting from long-term irradiation may increase the chances of colonic stem cell transformation.

There will be the following three projects addressing the impact of the following: 1) Space radiation on mouse models of intestinal tumorigenesis 2) Space radiation on neoplastic events in normal human colonocytes 3) Space radiation on GI stem cells (beginning in year 3)

**Rationale for HRP Directed Research:**

**Research Impact/Earth Benefits:**

Gastrointestinal tumors are frequent in the U.S, and the incidence of CR cancer in 2005 (latest year for which statistics are available) was 56.4 per 105 with a death rate of 21.0 per 105 or about 53,005 deaths per year. Colorectal cancer when considered in men and women is the second leading cancer killer in the United States ( http://www.cdc.gov/ ). Considering the high frequency of CR tumors in the American population, an even small increase by space radiation could have a major impact on risk estimates and planning of future manned space missions. The effect of complex DNA lesions, which are produced by HZE radiation, on intestinal tumorigenesis may provide insight into mutagenic processes affecting genome integrity and CR carcinogenesis.

Task Progress on Mouse Models of Colon Cancer:
Considering the increased colorectal cancer incidence in A-bomb survivors and the novel characteristics of space radiation, cancer causation by space radiation could potentially be even greater. One general approach for risk assessment is to determine the relative biologic effectiveness (RBE) of various parameters for space radiation compared to terrestrial radiation exposures. For HZE ions there is more uncertainty with RBE for cellular parameters, such as chromosome aberrations and cell lethality, where values obtained from in vitro studies typically ranges from 2 to 11-fold. Previously we have reported the RBE of proton (1.06) and 56Fe (1.25) for survival in female C57BL/6J mice. Here we report that in relation to gamma radiation the RBE factors of 28Si and 12C ions, which are also the major component of galactic cosmic radiation, are 1.41 and 0.99 respectively. The rationale for this study is that acute toxicity for the planned radiation sources at the proposed energies are not available in mice.

Initial pilot experiments with a solar particle event (SPE) simulation has been carried out using CDX2P Apc flox/+ mice as a model of colorectal adenoma-carcinoma progression to determine if the incidence or progression of colon cancer is increased after exposure to a total body of 2 Gy protons, a major component of space radiation. It will take about one year until we sacrifice the various cohorts of mice to determine if we observe an increase in the number of adenomas, adenoma-adenocarcinoma transitions, or the development of metastatic disease. The early results from our pilot experiments indicate that both the incidence (number of polyps) and percent invasive adenocarcinoma increases one year after a solar particle simulation (SPE) event.

For the intestinal tumorigenesis study we used APC1638N/+ model for quantitative and qualitative study on intestinal tumorigenesis. Previously we reported that exposure to 4 Gy 56Fe radiation led to significant increase in intestinal tumor numbers (compared to gamma radiation) in these mice. Qualitative analysis shows that exposure of APC1638N/+ mutant mice to 1 GeV/n Fe ions induces progression of colonic adenomatous lesions to adenocarcinomas about 2.5 months after irradiation (4 Gy). Recently we completed intestinal tumor scoring and histology for APC1638N/+ that had received 0.8 Gy of 1 GeV/n 56Fe radiation (equitoxic to 1 Gy of gamma rays) in a single exposure. Here we report that the frequency of colonic tumors was significantly enhanced in male mutants, but not in female mice. Furthermore, male

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| **Task Progress:** | and female control mice in our cohort – including moribund controls - never developed adenomas in the distal colon, while irradiated APC1638N/+male mice displayed in this region the highest relative increase for colonic tumor burden. No distal colonic adenomas were recorded in Apc1638N/+ female mice employed in our study. Importantly, we also observed that APC1638N/+ develop gastric adenomas after exposure to ionizing radiation. The presence of gastric tumors in APC1638N/+ mice has not been documented to date. We found that all three radiation types - gamma rays, 1 GeV/n Fe ions and 1 GeV/n protons – induced de novo gastric adenomas, which were classified as either fundic polyps or adenomas of the cardia, depending on their location.<br><br>It is now well established that colorectal cancer (CRC) develops as a consequence of the transformation of normal colonic epithelium to colon adenocarcinoma due to the progressive accumulation of genetic and epigenetic changes. We investigated space radiation induced signaling pathway activation in previously irradiated tissue samples obtained from APCMin/+ mice exposed to 5 Gy of gamma radiation and equitoxic doses of proton (4.7 Gy) and 56Fe radiation (4 Gy). Here we report that equitoxic doses of space radiation-induced enhanced expression of beta-catenin, which were observed by immunohistochemistry as well by western blot analysis.<br><br>Task Progress on Human Colonic Epithelial Cells:<br><br>1. Irradiation induced DNA damage in colonic epithelial cells is more difficult to repair when caused by iron particles compared to protons and gamma rays.<br><br>2. Iron damage persists longer in both 2D and 3D colonic epithelial cell cultures compared to protons and gamma rays.<br><br>3. Epithelial cells in intact human colonic tissues display more damage foci than surrounding stromal cells after both high and low LET radiations. All cells in intact colonic tissues are exposed to the same radiation dosage and are likely experiencing equal amounts of double strand breaks, but location in the cell cycle may explain why the replicating epithelial cells show more pronounced DNA damage response compared to quiescent stromal cells.<br><br>4. Additional cells from another patient have been established and characterized. We have an immortalized normal diploid cell line and one that is immortalized with one chromosomal change (trisomy 7). Trisomy 7 is present in about 65% of all colonic adenoma and adenocarcinomas. We have now progressed these cells partially toward tumorigenicity by reducing p53 expression (shRNA to p53), overexpressing K-ras and c- myc and are currently testing if irradiation converts these cells to cancer forming cells.<br><br>5. We have now demonstrated and published (Gastroenterology, 138:1012-1021, 2010) the ability to differentiate the immortalized human colonic epithelial cell lines. We showed that our human colonic epithelial cells express several gastrointestinal putative stem cell markers including CD44, CD29, BMI-1, and LGR5. This indicates that our immortalized human colonic epithelial cells can make similar 3D differentiated structures as do primary cells from mouse small intestine. Our cells are clonal, can divide long-term (are immortalized), and single cells divide to generate self-organizing structures with several differentiated cell types and architecture of crypts present in adults in the absence of a cellular niche. Our cells demonstrate polarized villin staining, junction formation (ZO-1), a subset of cells express Muc2 similarly to the staining of mouse small intestinal cells, and some even express chromogranin A (a neuroendocrine marker). This also helps demonstrate that we are culturing human colonic epithelial stem cells.<br><br>6. We have also passaged the cells with trisomy 7 (contain hTERT, cdk4, shRNAp53 and mutant K-ras) and observed that a subclone of these cells contain a trisomy 20 (also very common in colon cancer). We have also observed a non-random trisomy 20 in a HCECs + 7 without addition of mutant K-ras and shRNA to p53. While these cells make large colonies and grow anchorage independent in soft agar they do not make tumors in nude mice. We have now expressed c-myc in these cells and these also do not make tumors. These isogenic partially progressed cell lines are extremely useful for future low dose radiation experiments to determine biological steps in cancer progression and new experiments are in progress. |

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| **Bibliography Type:** | Description: (Last Updated: 06/30/2026) |
| **Abstracts for Journals and Proceedings** | Eskiocak U, Kim SB, Roig AI, Kitten E, Batten K, Cornelius C, Zou YS, Wright WE, Shay JW. "CDDO-Me Protects Against Heavy Ion-Induced Transformation of Human Colonic Epithelial Cells." NASA Human Research Program Investigators' Workshop, February 3-5, 2010, Houston, TX. Jerry W. Shay, Session chair, Space Radiation I, and oral presentation.<br>NASA Human Research Program Investigators' Workshop, February 3-5, 2010, Houston, TX.<br>http://www.dsls.usra.edu/meetings/hrp2010/pdf/Radiation/1103Shay.pdf , Feb-2010 |
| **Abstracts for Journals and Proceedings** | Trani D, Datta K, Doiron K, Fornace AJ Jr. "Space Radiation and Intestinal Tumorigenesis Risk Assessment." NASA Human Research Program Investigators' Workshop, February 3-5, 2010, Houston, TX. Al Fornace. Space Radiation II, (oral presentation).<br>NASA Human Research Program Investigators' Workshop, February 3-5, 2010, Houston, TX.<br>http://www.dsls.usra.edu/meetings/hrp2010/pdf/Radiation/1047Fornace.pdf , Feb-2010 |
| **Abstracts for Journals and Proceedings** | Kim SB, Eskiocak U, Cornelius C, Ly P, Biglione S, Shay JW. "Bardoxolone Methyl: A Novel Radiation Countermeasure and Mitigator." 18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. (oral presentation).<br>18th IAA Humans in Space Symposium, Houston, TX, April 11-15, 2011. , Apr-2011 |
| **Abstracts for Journals and Proceedings** | Eskiocak U, Kim SB, Ly P, Biglione S, Komurov K, Cornelius C, Wright WE, White MA, Shay JW, Roig AI. "Stable Down-Regulation of Putative Colorectal Cancer Candidate Genes Identifies Extensive Fragility in the Matrix-Dependent Proliferation Control Network." Poster presented at DDW (Digestive Disease Week) 2011 Meeting, Chicago, IL, May 7-10, 2011.<br>DDW (Digestive Disease Week) 2011 Meeting, Chicago, IL, May 7-10, 2011. (poster received distinction status).<br>http://ddw.scientificposters.com/epsabstract.cfm?id=1 , May-2011 |

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| **Abstracts for Journals and Proceedings** | Trani D, Datta K, Moon B-H, Williams E, Kallakury B, Fornace AJ Jr. "Radiation quality and enhancement of intestinal tumor multiplicity and grade in vivo." Colon Cancer in Murine Models and Humans III, Bar Harbor Club, Bar Harbor, Maine, USA, October 18-22, 2010.<br>Colon Cancer in Murine Models and Humans III, Bar Harbor Club, Bar Harbor, Maine, USA, October 18-22, 2010. , Oct-2010 |
| **Abstracts for Journals and Proceedings** | Trani D, Moon B-H, Kallakury B, Hartmann D, Datta K, Fornace AJ Jr. "Dependent Differences in Intestinal Tumorigenesis Induced in Apc1638N/+ Mice by low- and high-LET Radiation." 14th International Congress of Radiation Research, Warsaw, Poland, August 28-September 1, 2011.<br>14th International Congress of Radiation Research, Warsaw, Poland, August 28-September 1, 2011. , Aug-2011 |
| **Abstracts for Journals and Proceedings** | Trani D, Moon B-H, Kallakury B, Datta K, Li H, Williams EM, Fornace AJ Jr. "Quantitative and Qualitative Features of Intestinal Tumorigenesis Induced in Apc1638N/+ Mice by Low- and High-LET Radiation." 22nd Annual NASA Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011.<br>22nd Annual NASA Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. http://www.dsls.usra.edu/meetings/radiation2011/pdf/7050.pdf , Sep-2011 |
| **Abstracts for Journals and Proceedings** | Li H, Wang Y, Zhou B, Trani D, Ressom H, Fornace AJ Jr. "Characterization of the effects of gamma and proton radiation on immune cells using genetic and metabolomics approaches." 22nd Annual NASA Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011.<br>22nd Annual NASA Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. http://www.dsls.usra.edu/meetings/radiation2011/pdf/7092.pdf , Sep-2011 |
| **Abstracts for Journals and Proceedings** | Suman S, Datta K, Trani D, Fornace AJ Jr. "Heavy ion radiation induces preferential activation of ß-catenin/TCF-4 signaling in APCMin/+ intestinal tumors." 22nd Annual NASA Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011.<br>22nd Annual NASA Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. http://www.dsls.usra.edu/meetings/radiation2011/pdf/7095.pdf , Sep-2011 |
| **Abstracts for Journals and Proceedings** | Datta K, Suman S, Trani D, Laiakis EC, Strawn SJ, Fornace AJ Jr. "RBE for lethality following exposure to 12C and 28Si radiation." 22nd Annual NASA Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011.<br>22nd Annual NASA Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. http://www.dsls.usra.edu/meetings/radiation2011/pdf/7091.pdf , Sep-2011 |
| **Abstracts for Journals and Proceedings** | Suman S, Datta K, Trani D, Fornace AJ Jr. "Space radiation-induced intestinal tumorigenesis is mediated via preferential activation of ß-catenin/TCF-4 signaling." 14th International Congress of Radiation Research, Warsaw, Poland, August 28-September 1, 2011.<br>14th International Congress of Radiation Research, Warsaw, Poland, August 28-September 1, 2011. , Aug-2011 |
| **Articles in Peer-reviewed Journals** | Ly P, Eskiocak U, Kim SB, Roig AI, Hight SK, Lulla DR, Zou YS, Batten K, Wright WE, Shay JW. "Characterization of aneuploid populations with trisomy 7 and 20 derived from diploid human colonic epithelial cells." Neoplasia. 2011 Apr;13(4):348-57. PubMed PMID: 21472139 , Apr-2011 |
| **Articles in Peer-reviewed Journals** | Eskiocak U, Kim SB, Ly P, Roig AI, Biglione S, Komurov K, Cornelius C, Wright WE, White MA, Shay JW. "Functional parsing of driver mutations in the colorectal cancer genome reveals numerous suppressors of anchorage-independent growth." Cancer Res. Priority Reports, 2011 Jul 1;71(13):4359-65. Epub 2011 Apr 28. PubMed PMID: 21527559 , Jul-2011 |
| **Articles in Peer-reviewed Journals** | Wang H, Venkatesh M, Li H, Goetz R, Mukherjee S, Biswas A, Zhu L, Kaubisch A, Wang L, Pullman J, Whitney K, Kuro-O M, Roig AI, Shay JW, Mohammadi M, Mani S. "Pregnane X receptor activation induces FGF19-dependent tumor aggressiveness in humans and mice." Journal of Clinical Investigation. 2011 Aug;121(8):3220-32. Epub 2011 Jul 11. PubMed PMID: 21747170 ; http://dx.doi.org/10.1172/JCI41514 , Aug-2011 |
| **Awards** | Eskiocak U, Kim SB, Ly P, Biglione S, Komurov K, Cornelius C, Wright WE, White MA, Shay JW, Roig AI. "Poster of Distinction for 'Stable Down-Regulation of Putative Colorectal Cancer Candidate Genes Identifies Extensive Fragility in the Matrix-Dependent Proliferation Control Network', Digestive Disease Week 2011, Chicago, IL, May 7-10, 2011. http://ddw.scientificposters.com/epsabstract.cfm?id=1 " May-2011 |